

ECON 108

INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

Money, banking, and central banking

CHAPTER VIII (from the book)

CHAPTER OUTLINE

The Functions of Money

Properties of Money

Defining Money

The Money Multiplier

What is Money?

Money

- Any medium that is universally accepted in an economy both by sellers of goods and services as payment for those goods and services and by creditors as payment for debts

What is Money?

Iron	Boar tusk	Playing cards
Copper	Red woodpecker scalps	Leather
Brass	Feathers	Gold
Wine	Glass	Silver
Corn	Polished beads (wampum)	Knives
Salt	Rum	Pots
Horses	Molasses	Boats
Sheep	Tobacco	Pitch
Goats	Agricultural implements	Rice
Tortoise shells	Round stones with centers removed	Cows
Porpoise teeth	Crystal salt bars	Paper
Whale teeth	Snail shells	Cigarettes

Source: Roger LeRoy Miller and David D. VanHoose, *Money, Banking, and Financial Markets*, 3rd ed. (Cincinnati: South-Western, 2007), p. 7.

What is Money?

Money (a stock concept) is different from:

Wealth: the total collection of pieces of property that serve to store value

Income: flow of earnings per unit of time
(a flow concept)

Functions of Money

The functions of money

- Medium of exchange
- Unit of accounting
- Store of value (purchasing power)
- Standard of deferred payment

Functions of Money

Medium of Exchange:

- Any item that sellers will accept as payment
- Money facilitates exchange by reducing transaction costs associated with means-of-payment uncertainty
 - Permits specialization, facilitates efficiencies

Functions of Money

A Medium of Exchange must:

- be easily standardized
- be widely accepted
- be divisible
- be easy to carry
- not deteriorate quickly

Functions of Money

Barter

- The direct exchange of goods and services for other goods and services without the use of money
- Simply a direct exchange requires a *double coincidence of wants*

Functions of Money

Unit of Accounting

- A measure by which prices are expressed
- The common denominator of the price system

Functions of Money

Store of Value

- The ability to hold value over time
- Money allows you to transfer value (wealth) into the future

Functions of Money

Standard of Deferred Payment

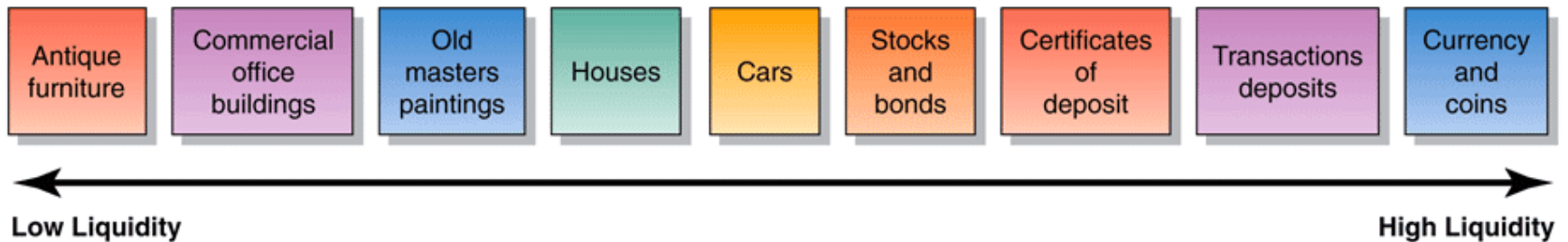
- A property of an item that makes it desirable for use as a means of settling debts maturing in the future

Properties of Money

Liquidity

- The degree to which an asset can be acquired or disposed of without much danger of any intervening loss in *nominal* value and with small transaction costs
- Money is the most liquid asset

Degrees of Liquidity



Properties of Money

Question

- What is the cost of holding money (its opportunity cost)?

Answer

- It is the alternative interest yield obtainable by holding some other asset

Properties of Money

Questions

- What backs money?
- Is it gold, silver, or the federal government?

Answer

- Your confidence

Properties of Money

Fiduciary Monetary System

- A system in which currency is issued by the government and its value rests on the public's confidence that it can be exchanged for goods and services
- The Latin *fiducia* means “trust” or “confidence”

Properties of Money

Transactions Deposits

- Checkable and debitable account balances in commercial banks and other types of financial institutions, such as credit unions and mutual savings banks
- Any accounts in financial institutions on which you can easily transmit debit-card and check payments without many restrictions

Properties of Money

Currency and transactions deposits are money because of their

- Acceptability
- Predictability of value

Defining Money

Money is important

- Changes in the rate at which the money supply increases or decreases affect important economic variables (at least in the short run) such as inflation, interest rates, employment, and the level of real GDP

Money Supply

- The amount of money in circulation

Defining Money

Economists use two basic approaches to define and measure money

- The transactions approach
- The liquidity approach

Defining Money

Transactions Approach

- A method of measuring the money supply by looking at money as a medium of exchange

Liquidity Approach

- A method of measuring the money supply by looking at money as a temporary store of value

Defining Money

The transactions approach to measuring money: **M1**

- Currency and coins
- Transactions (checkable) deposits
- Traveler's checks

Defining Money

MI

- Currency
 - Minted coins and paper currency not deposited in financial institutions
 - The bulk of currency “in circulation” actually does not circulate within the U.S. borders

Defining Money

MI

- Transactions deposits
- Any deposits in a **thrift institution** or a commercial bank on which a check may be written or debit card used

Thrift Institution

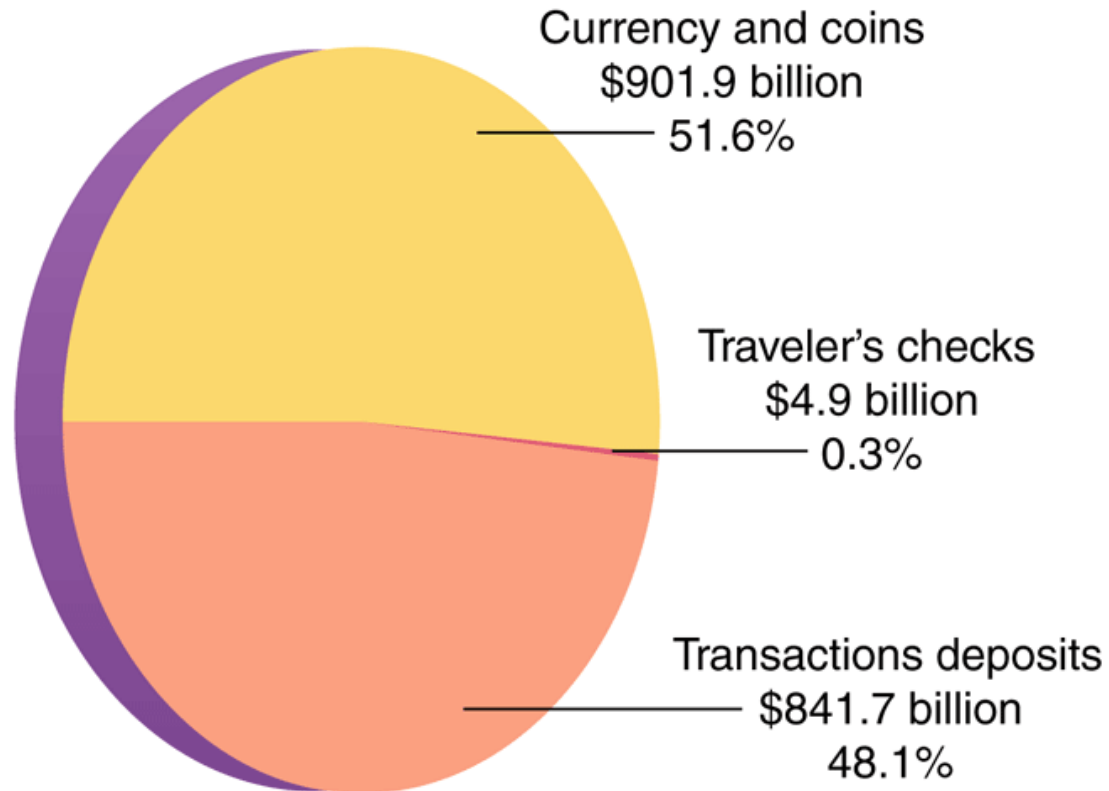
- Financial institutions that receive most of their funds from the savings of the public

Defining Money

MI

- Traveler's Checks
- Financial instruments purchased from a bank or a nonbanking organization and signed during purchase that can be used as cash upon a second signature by the purchaser

Panel (a)



M1 = \$1,748.5 billion

Defining Money

The liquidity approach to measuring money: **M2**

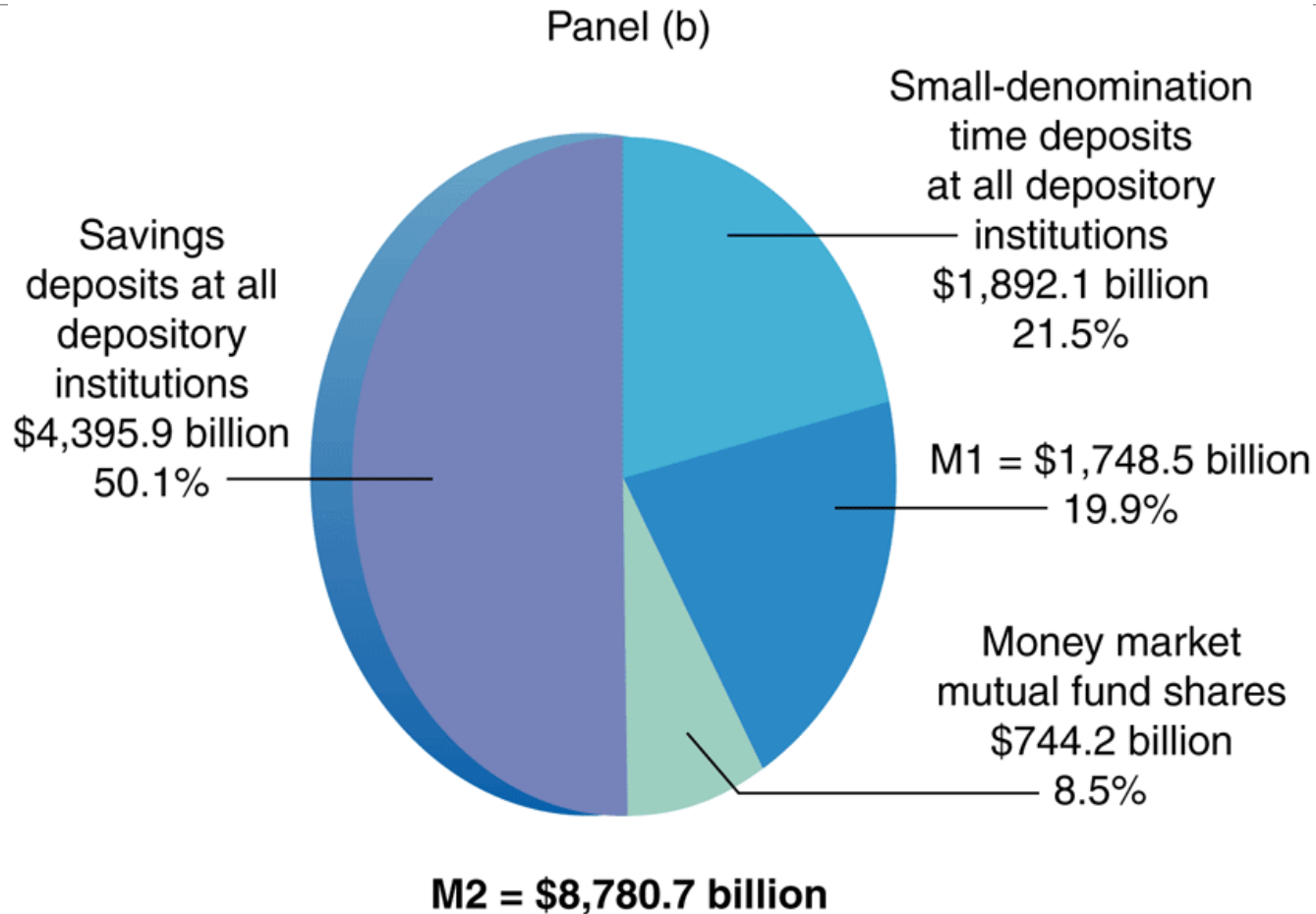
- Assets that are almost money
- Highly liquid
- Easily converted to cash
- Time deposits are an example

Defining Money

The liquidity approach: **M2** is equal to M1 plus

1. Savings deposits
2. Small denomination (<\$100,000) time deposits
3. Balances in retail money market mutual funds

M2



Sources: Federal Reserve Bulletin; Economic Indicators, various issues; author's estimates.

The Money Multiplier

Fractional Reserve Banking

- A system in which depository institutions hold reserves that are less than the amount of deposits
- Originated when goldsmiths in Greece issued notes that exceeded the value of gold and silver on hand

The Money Multiplier

Required reserves

Reserves that the Fed requires banks to hold. The fraction of its deposits that banks are required to hold is called the required reserve ratio. Currently, the Fed pays no interest on reserves.

Excess reserves

Any additional reserves the banks choose to hold (excess reserves).

The Money Multiplier

Money Creation by the Banking System

Assume:

1. Reserve ratio is 10 percent
2. Zero excess reserves are kept
3. Transactions deposits are the bank's only liabilities and loans are the bank's assets
4. Every time a loan is made, the proceeds are put into a deposit account, i.e., individuals do not keep cash.

The Money Multiplier

Money Creation by the Banking System

What is the effect on total deposits and the money supply?

Money Multiplier

- A number that, when multiplied by a change in reserves in the banking system, yields the resulting change in the money supply

The Money Multiplier

Potential money multiplier

- The reciprocal of the required reserve ratio, assuming no leakages into currency and no excess reserves
- It is equal to 1 divided by the required reserve ratio

$$\text{Potential money multiplier} = \frac{1}{\text{Reserve ratio}}$$

The Money Multiplier

The actual money multiplier usually is smaller than the potential money multiplier because:

Banks may hold excess reserves.

Households may keep cash out of banks.

$$\begin{array}{ccccc} \text{Actual change} & & & & \\ \text{in the money} & & & & \\ \text{supply} & = & \text{Actual money} & \times & \text{Change in} \\ & & \text{multiplier} & & \text{total reserves} \end{array}$$

The Money Multiplier

Real-world money multipliers

- M1 multiplier: 1.5 to 2.0
- M2 multiplier: 6.5 in the 1960s to over 12 in the mid-2000s; about 5 since then

The Money Multiplier

$$m = \frac{1 + c}{r + c + e}$$

c: is the ration of cash kept by households to deposits

r: is the reserve ratio

e: is the ratio of excess reserves to deposits

Example

$r = \text{required reserve ratio} = 0.10$

$C = \text{currency in circulation} = \$400B$

$D = \text{checkable deposits} = \$800B$

$ER = \text{excess reserves} = \$0.8B$

$M = \text{money supply (M1)} = C + D = \$1,200B$

$$c = \frac{\$400B}{\$800B} = 0.5$$

$$e = \frac{\$0.8B}{\$800B} = 0.001$$

$$m = \frac{1 + 0.5}{0.1 + 0.001 + 0.5} = \frac{1.5}{0.601} = 2.5$$

This is less than the simple deposit multiplier
Although there is multiple expansion of deposits,
there is no such expansion for currency

END OF CHAPTER 8